

Emotional Priming Effects on the Interpretation of Ambiguous Images

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Introduction

(Written by: E. Homar, E. Huebner, L. Doll)

Previous research has found evidence that emotional stimuli can trigger affective priming, or the influence of affective states on the accessibility of information. This process involves earlier emotional encounters influencing how the current stimulus is perceived (Martinez et al., 2023). One of the seminal studies regarding affective priming was Murphy and Zajonc's 1993 paper demonstrating that both positive and negative reactions can be induced with minimal exposure and cognitive processing (Murphy & Zajonc, 1993). This finding supported Zajonc's affective priming hypothesis, suggesting that emotional stimuli may be processed at a faster rate than conscious, cognitive processing. However, this hypothesis has been contested. While the affective priming hypothesis suggests that the priming occurs outside of conscious awareness, other studies support the idea that these cognitive processes are actually intertwined with the emotional priming. A study utilizing Murphy and Zajonc's same experimental paradigm found that longer exposure to affective facial expressions resulted in priming, but not when the levels of exposure were suboptimal; this demonstrated that affective priming must involve at least some level of cognitive processing (Kemp et al., 1996). This later finding aligns with Joseph LeDoux's theory of affective-cognitive interaction, which states that emotion and cognition are controlled by separate but intertwining systems in the brain (LeDoux, 1998).

The affective-cognitive interaction theory is supported by both anatomical and neural substrates of the brain. The amygdala has traditionally been heavily linked with emotional responses, such as fear (LeDoux, 1998). Recent research has built on and advanced these findings by demonstrating that the amygdala is not an isolated structure, but also part of a broader network that interacts with cognitive systems involved in attention, memory, and decision making (Pessoa, 2008). For example,

connections between the amygdala and prefrontal cortex allow emotional information to influence cognitive processing, while cognitive control processes can regulate emotional responses (Gross, 2002). Research on cognitive-emotional interactions of the brain's neural networks primarily has focused on fluctuations in fMRI signals that demonstrate communication between these anatomical regions (Andellini et al., 2015). Along with this, interactions between the amygdala and hippocampus demonstrate how emotional arousal enhances memory encoding, especially for emotionally salient stimuli (Cahill & McGaugh, 1998). This finding speaks to one of the most essential dialogues for emotional-cognitive signaling, modulation of attention and memory. Specifically, fear-based emotions such as stress and anxiety have been shown to influence attention and working memory (Aue & Okon-Singer, 2015). This theme of modulating cognitive control by the infusion of negative emotions is a theme we continue to investigate in our present research.

Furthermore, prior studies have found that people react differently to ideas of meaning in ambiguous images, and negative priming may be associated with unconscious decisions on their meanings (Filippova & Kostina, 2019). Furthermore, when primed with emotional words, participants show faster and more accurate identification of a facially portrayed emotion when the emotion and primed word match categorically, compared to when the emotional word and face are mismatched (Carroll & Young, 2005). Thus, emotional category words such as “happy” have been shown to elicit priming effects and influence interpretations of facial images, a concept we continue to explore. However, here we are instead interested in how this observed emotional priming maps onto ambiguous facial images rather than faces with specific emotional portrayals.

Research consistently shows that humans demonstrate increased salience for negative compared to positive entities. This phenomenon, coined as negativity bias, exists across many facets of life including emotions, events, and personality traits (Rozin & Royzman, 2001). Some research argues for a larger finding that bad is simply stronger than good, as negativity bias is still enacted even if levels of saliency are controlled for (Baumeister, 2001). Predictably, this tendency is increased in individuals who suffer from clinical depression, as an overactivated amygdala

selectively attunes to negative emotions and rumination (Jiang, 2024). Healthy, non-depressed participants who had equal accessibility to both negative and positive emotions demonstrated negativity bias that was positively correlated with activation in the region of the bilateral pregenual anterior cingulate cortex (pgACC) when presented with ambiguous facial stimuli. This means that when ambiguous faces were perceived as sad instead of happy, there was increased activation in the pgACC region (Ito, 2017). Our research extends these findings by observing how negativity bias plays out when accessibility of positive or negative emotion is influenced by our priming measures. Additionally, it could potentially provide support for the theory that bad emotions are simply “stronger” than positive ones, as all of our conditions will have an equivalent amount of priming with the same number of words shown for each condition.

Accordingly, we plan to investigate the priming effect of emotion-inducing words on ambiguous images. By comparing the responses of participants primed with positive emotion-inducing words, negative emotion-inducing words, and neutral words, we aim to clarify how different types of emotions influence the strength of the priming effect. We use the Thematic Apperception Test (TAT) in order to do this as past studies have found that affective tone from previous picture cards can influence responses to the test, suggesting the presence of motivational priming (Bozza, 2017). Although concerns of the validity and reliability of the TAT for clinical purposes have been brought into question, the test’s ambiguity remains uncontested and fits our goal of demonstrating how priming effects may act on images that are ambiguous (Miller, 2015).

Hypothesis

(Written by: E. Homar, E. Huebner, L. Doll)

We hypothesized that both positive and negative emotion-inducing words will have a stronger priming effect compared to non-emotional or neutral words, measured by their effect on the participant’s image interpretations. Between the two categories of emotional words, we expect that negative words will have a stronger priming effect compared to the positive words, aligning with previous research and theories regarding negativity bias.

Methods

(Written by: E. Homar, E. Huebner, L. Doll)

Our experiment was a between-subjects design. Each participant was randomly assigned to one of three groups when they began the task: positive emotion, negative emotion, or neutral. The neutral condition acted as our control. Participants were assigned to only one of the three categories and received words from only that category in order to prevent potential crossing over of priming, as well as minimize demand characteristics. Participants were adults from Wisconsin recruited by text message. Our sample size was 30 participants, as this number is often used as a common rule-of-thumb minimum based on central limit theorem logic that a sample of this size usually achieves normally distributed data. Additionally, we found this number to be an attainable target for our purposes.

We used the Open AI platform Google Gemini to write code for our web app. Through a Google Apps Script, this code communicated with a Google Sheet to populate our database with participants' responses. We told Google Gemini to create a HTML file that portrayed a Welcome screen with instructions for our task and a "Start" button that assigned them a random condition and participant ID. This took them to complete the task design as we describe below, first walking them through the priming stage and then the image interpretation stage. The timestamp, participant ID, condition, and all participant responses were recorded through the connected Google sheet. GitHub was used to deploy our coded HTML file on the web so that it could be shared and completed by participants.

After pressing "Start," each participant was primed with a list of ten words. Based on their randomly assigned condition, these words had either a positive emotion-inducing theme (joy, laughter, victory, paradise, thrill, kindness, freedom, beloved, radiant, triumph) a negative emotion-inducing theme (agony, failure, poison, poison, grief, danger, hatred, sickness, tragedy, shame, guilt), or neutral words (chair, handle, pencil, curtain, concrete, avenue, standard, platform, utensil, adjacent). To ensure that participants encoded each word, participants were presented with the words one at a time and asked to type the word they were shown in a text box below it. Following the priming stage, participants in each group assessed five different

ambiguous images from the Thematic Apperception Test (TAT). Participants viewed these images one at a time in a random order and were prompted to write what emotions they believed were evident in the image by the phrase, “List the emotions seen in this image.” There was a maximum word count of three words on the response in order to prevent a lengthy response that included too many emotions to easily code. Following this, the participants were also prompted to tell a brief story of what was happening in the image, using the words that they had provided in the previous prompt.



Figure 1. Example of an ambiguous image from the Thematic Apperception Test (TAT) presented to participants, who were asked to identify perceived emotions and write a brief narrative based on the image.

We coded the “list the emotions seen in this image” responses for each image as neutral, positive, or negative based on its content. A positive word received a score of +1, a negative word received a score of -1, and a neutral word received a score of 0. Scores were totaled for each participant, then averaged across each test group. Responses were compared across subjects for our experimental design using a one-way ANOVA.

Results

(Analyzed by E. Huebner)

A one-way analysis of variance (ANOVA) was conducted to examine the effects of emotional priming condition (negative, positive, neutral) on participants’ interpretation scores of ambiguous facial images. The analysis revealed no significant differences between the three experimental groups, $F(2, 22) = 1.94$, $p = .168$.

Although the overall effect was not statistically significant, descriptive patterns suggested variation across conditions. Participants in the negative word condition

appeared to demonstrate a greater tendency towards negative interpretations compared to participants in the positive and neutral conditions. This positive word group showed a tendency towards positive conditions relative to the neutral group, but this effect was less pronounced than the pattern observed in the negative condition.

Because the ANOVA did not reach statistical significance, these observed differences should be interpreted with caution, as they may reflect random variation rather than a reliable effect of priming condition. No post hoc comparisons were conducted, given the original non-significant test.

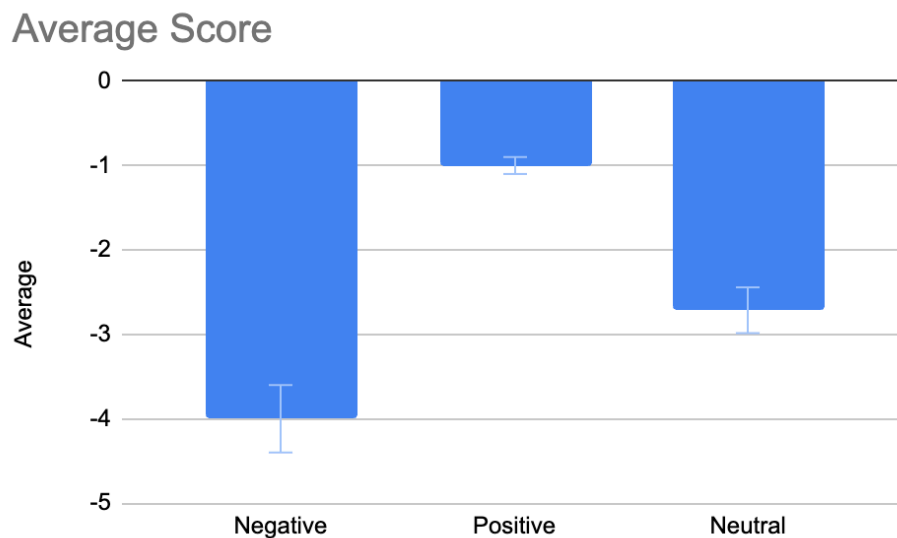


Figure 2. Average emotional interpretation scores across priming conditions (negative, positive, and neutral). Error bars represent ± 1 standard error of the mean. Participants in the negative condition showed a greater tendency towards negative interpretations relative to the positive and neutral conditions, while the positive condition showed an increase in positive interpretations compared to the neutral condition. However, these differences were not statistically significant. $F(2,22) = 1.94$, $p = .168$.

Discussion

(Written by: E. Homar, E. Huebner, L. Doll)

Our study examined whether emotional priming influences the interpretation of ambiguous images and whether negative emotional stimuli produce a stronger priming effect than positive stimuli. Contrary to our hypothesis, the results did not reveal a

statistically significant difference between priming conditions. Although participants in the negative condition appeared to show a greater tendency toward negative interpretations, and participants in the positive condition showed a shift towards positive interpretations relative to the neutral group, these patterns did not reach significance and therefore cannot be interpreted as reliable effects.

Despite the lack of statistical significance, the observed descriptive trends are consistent with prior literature suggesting that emotionally valenced stimuli can bias perception. In particular, the stronger shift observed in the negative condition aligns with research on negativity bias, which proposes that negative information has a greater influence on cognition than positive information (Rozin & Royzman, 2001). Additionally, the result that every single condition (including the positively primed participants) produced a negative score supports this phenomenon as well. Regardless of the affective priming the participants experienced, negative stories, events, and emotions may have prevailed with increased saliency. However, because of the lack of statistical significance, this effect should be viewed with skepticism. One possible interpretation is that emotional priming and negativity bias may still influence perception, but the effect size in this study was too small to be detected, given the current sample.

Several limitations of the study might explain the lack of statistically significant findings. First, the sample size was relatively small, which likely reduced statistical power and made it more difficult to detect true differences between groups. Second, participants were recruited through convenience sampling (friends, family, and classmates), which may limit generalizability and introduce bias. Another potential limitation is the presence of demand characteristics. Participants may have inferred the purpose of the study and adjusted their responses to align with the emotional tone of the words they were presented, potentially inflating the observed trends. Future studies should address these limitations by increasing sample size to improve statistical power and using more diverse recruitment methods to increase generalizability. Another potential rationale for the insignificance produced by our experiment may be a weak manipulative structure. For instance, perhaps the introduction of a higher volume of the affective priming words would have resulted in a stronger effect. Additionally, past research has shown that emotional information only maintains influence if attended to

and prioritized for participants, thus not having effects when irrelevant to a task at hand (Mauersberger et al., 2025). This may be another possible explanation, as perhaps the word priming task was not a pertinent enough manipulation for participants to be attentive to it. Previous research also suggests that, in congruence with LeDoux's emotion-cognition theory, the effects of emotional priming depend on the degree of conscious experience (Lohse & Overgaard, 2019). Although our word priming task involved participants typing the word and thus holding it consciously, it is possible that the level of awareness was insufficient to have the lasting effects we were seeking.

In conclusion, while the present study did not find statistically significant evidence for affective priming effects on the interpretation of ambiguous images, the observed patterns suggest that emotional salience, especially negative, may still influence perception. If, with the proposed changes to improve the sampling methods, the result was pushed to statistical significance, it would align and support previous literature on this topic. With the introduction of a larger and more unbiased sample, the trend within our data may be able to reach a reliable effect supporting past research done on affective priming, emotion-cognitive interactions, and negativity bias.

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